

Attention Is All You Need

CS795 Presentation 1

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Exploring the architecture that transformed Natural Language Processing and paved the way for LLMs.

The Paper

Vaswani, A., Shazeer, N., Parmar, N., et al. 2017. *Attention is All You Need*. NeurIPS 2017.
[arXiv:1706.03762](https://arxiv.org/abs/1706.03762)

- ➡ Moved away from RNNs and LSTMs
- ➡ Laid foundation for LLMs like GPT

Objectives

- Introduce the **Transformer** architecture
- Explain **self-attention** and **multi-head attention**
- Highlight **key results** and why this paper changed NLP
- Discuss **strengths, limitations, and future directions**

Motivation

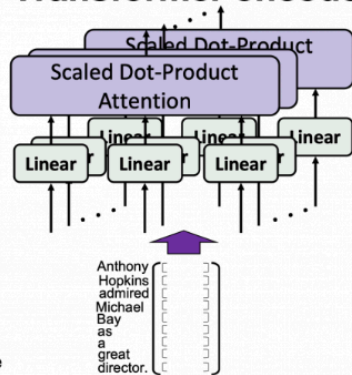
- RNNs and LSTMs process input **sequentially** → slow and hard to parallelize
- Struggle with **long-range dependencies**
- CNNs help but require very deep stacks for global context
- **Goal:** Design an architecture that is faster, scalable, and captures dependencies across entire sequence efficiently

■ RNN encoder



- In RNN encoders, you have to wait for all the time steps to encode input sentences.
- Especially gated RNNs need a lot of nonlinear transformations.
- Also, RNN is potentially subject to vanishing gradient problems.

■ Transformer encoder



- In Transformer encoders, you can put one input sentence at once.
- Any tokens can attend to the other tokens at once, on various standards.

Key Idea

"Attention is all you need."

- Use **self-attention** to relate every token to every other token in parallel
- Add **positional encodings** to preserve word order
- Leverage **multi-head attention** to capture different types of relationships simultaneously

Transformer Architecture

- **Encoder:** Stacks of self-attention + FFN
- **Decoder:** Masked attention + cross-attention to encoder outputs
- **Residual + LayerNorm** stabilize training
- **Position-wise FFN** applied identically to each token

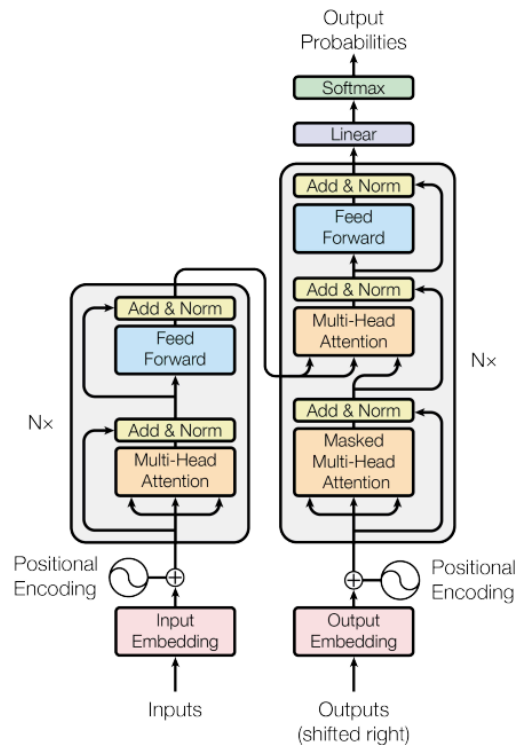
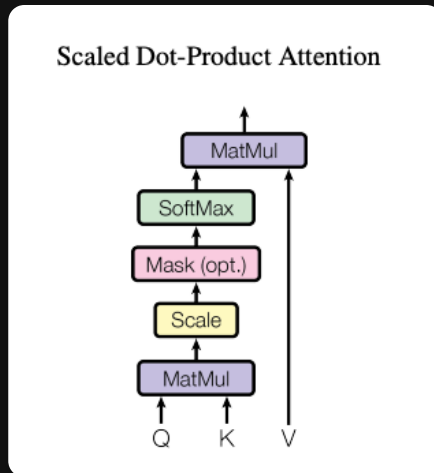


Figure 1: The Transformer - model architecture.

🔍 Self-Attention

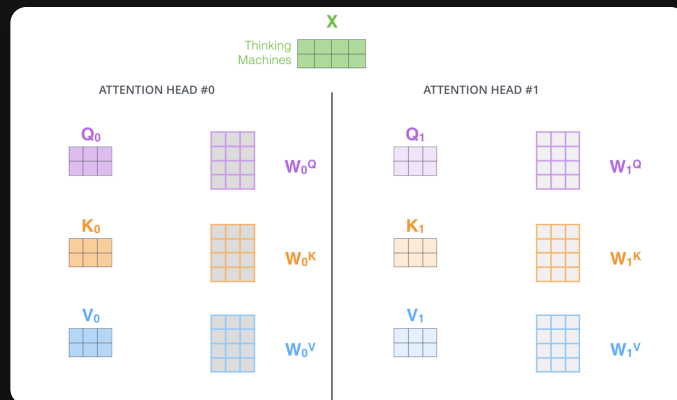
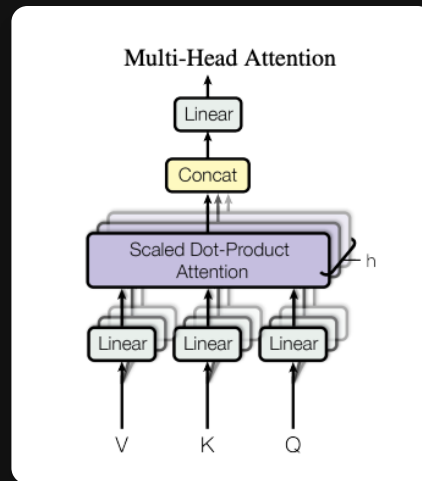
1. Compute similarity scores between tokens
2. Apply softmax to get attention weights
3. Combine values into context-aware representation

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$



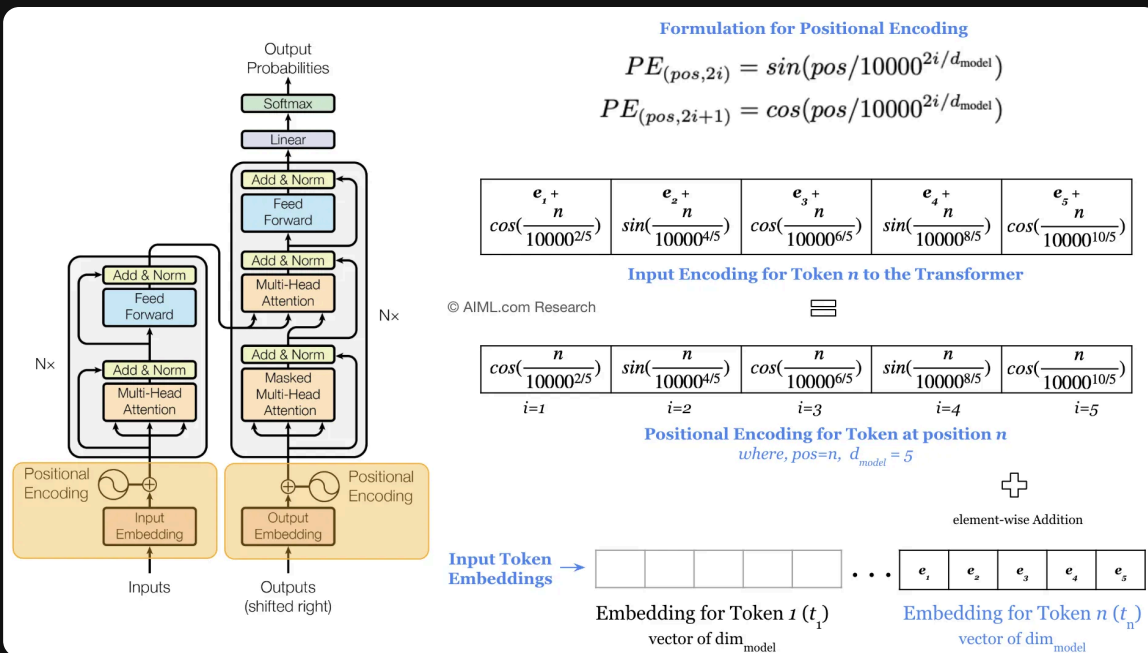
🎭 Multi-Head Attention

- **Multiple heads** learn diverse relationships
- Outputs concatenated and projected back
- Improves model expressiveness





Positional Encoding



- Encodes order with sin/cos patterns
- Allows extrapolation to longer sequences



Results (WMT 2014)

- **English→German:** 28.4 BLEU (SOTA)
- **English→French:** 41.8 BLEU (SOTA)
- **2× faster training** vs. best RNN models
- **First SOTA model** without recurrence or convolution

Strengths

- **Parallelizable** → faster training/inference
- **Handles long-range dependencies**
- **Simplicity**: no recurrence or convolution
- **Interpretability**: visualize attention weights
- Foundation for **BERT, GPT, T5, etc.**

Limitations

- **Quadratic complexity** $O(n^2)$ → compute + memory bottleneck
- Requires **large datasets + compute**
- Fixed **positional encoding** may not generalize to longer contexts

Conceptual Impact

- Sparked the **Transformer Revolution**
- Scaled with data/compute → emergence of LLMs
- Extended beyond text → Vision, Audio, Multimodal Transformers (e.g. ViT (Vision Transformer) or Whisper)

Follow-up Work

- **BERT (2018):** Bidirectional pretraining
- **GPT (2018+):** Autoregressive transformers
- **T5 (2019):** Unified text-to-text
- **Efficient Transformers:** Linformer, Performer, Longformer



Future Directions

- Sparse/Linear Attention → reduce compute
- Long-context models (1M+ tokens)
- Multimodal fusion (text+image+audio)
- Better interpretability + safety research
- Memory-efficient attention mechanisms like FlashAttention or xFormers

Key Takeaways

- Attention enabled **parallelization + scalability**
- Paved way for **LLMs**
- Efficiency + interpretability are active research frontiers

 Thank You!

Questions & Discussion